

Software Requirements Specification Template

Software Engineering

The following annotated template shall be used to complete the Software Requirements Specification (SRS) assignment.

Template Usage:

Text contained within angle brackets ('<', '>') shall be replaced by your project-specific information and/or details. For example, <Project Name> will be replaced with either 'Smart Home' or 'Sensor Network'.

Showtime Hub

Software Requirements Specification

Version 1

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Group #9

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Document Approval

The following Software Requirements Specification has been accepted and approved by the following:

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for ShowTime Hub, an online ticketing and concession-ordering website for the ShowTime Hub theater chain. It establishes the functionality and operational expectations the system must meet to support customers, theater staff, and administrators.

This document is intended for software developers, designers, testers, project managers, and theater management. It provides a shared reference for development decisions, system testing, and acceptance of the completed product.

1.2 Scope

ShowTime Hub will be a movie ticketing and concession-ordering platform accessible through a website and mobile application. Customers can browse current and upcoming movies, view movie details and reviews, and search for screenings by title, date, time, location, and viewing format. Listings will include special events and information about available services.

Customers will be able to view seat availability, select seats, purchase individual or group tickets, and receive electronic tickets with a unique identifier. They can also purchase concessions and receive a claim code for pickup. Customer accounts will support purchase history, saved payment methods, rewards, badges, notifications, and sharing showtime and seating information.

Authorized theater staff will manage movie listings, showtimes, pricing, concessions, and sales reports. System administrators will manage platform-wide settings and season themes. For selected films, the platform will display external streaming availability but will not stream the movie directly.

The goal of ShowTime Hub is to bring movie discovery, visit planning, and purchasing into one platform, reduce the need for box-office purchases, and give theater staff a central system for managing screening information and online sales.

<Showtime Hub>

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification: A document describing the system's required features and constraints.
API	Application Programming Interface: Allows ShowTime Hub to exchange information with other systems.
Showtime	A scheduled movie screening at a specific theater, data, and time.
Viewing Format	A scheduled movie screening at a specific theater, date, and time.
Seat map	A visual layout showing seats, availability, and designated accessible seatings.
E-Ticket	An electronic admission ticket with a unique identifier, viewable on one device or printable.
Concessions	Food and beverages available for purchase at a theater.
Claim code	A unique code used to identify a concession order during pickup.
Reward points	Credits earned through eligible purchases and redeemed according to program rules.
Badge	A digital achievement awarded for meeting specified movie-attendance criteria.

Closed captioning	Text displaying dialogue and relevant sounds through a supported device or display.
Audio Description	Spoken descriptions of important visual elements in a movie.
SMS	Short Message Service: text messaging used for customer notifications.
Theater Administrator	An authorized user who manages information and operations for assigned theater locations.
System Administrator	An authorized user who manages platform-wide settings, permissions, and context.

1.4 References

No external documents are incorporated by reference in this version of the ShowTime Hub SRS. As business policies and service providers are completed, this section will include applicable theater policies, payment-service documentation, movie-info sources, and notification-service documentation.

Each reference will identify the document's title, author or publishing organization, date or version where available, and the location from which it can be obtained.

1.5 Overview

Section 2 will provide a general description of ShowTime Hub, including its relationship to theater operations, major functions, user characteristics, constraints, and assumptions and dependencies.

Section 3 defines the specific requirements for the website and mobile application. It covers external interfaces. This includes movie discovery, ticket purchasing, concession ordering, and account management. It also covers use cases, classes, and objects. It also establishes requirements for performance, reliability, availability, security, maintainability, and portability, along with design constraints and database requirements.

2. General Description

2.1 Product Perspective

ShowTime Hub will be a new and improved system where it will replace the old functions of box office processing, such as ticket booths and phone reservations. This system can be compared to other similar concepts like the Regal app and Fandango. Where people can find movies and showtimes, seat availability, pay online through the app or website and at kiosks. The system also allows the user to acquire tickets electronically where they can print it or use it through their mobile devices. ShowTime Hub will be a product where it will be its own theatre chain.

2.2 Product Functions

ShowTime Hub will include functions that allow the customers to browse movies that are currently showing or ones that are upcoming. Customers can filter the movie viewing format to (regular, 3D, 4DX, IMAX), select preferred showtimes, select seats from a visual, real-time seating display which also highlights sections for wheelchair accessibility and companions. Tickets can be purchased individually or as a group in which the customer will receive a unique code, secured payment through credit/debit cards, and issue a printable e-ticket or used through their mobile devices. The product will also allow customers to view past purchases, opportunities to leave a review, and a loyalty/rewards program earned based on the number of movies watched. Theatre staff and administrators can update movie pricings, showtimes, and check sales reports. Product also allows other theatre companies to list their own movies, showtimes, and seat mappings on ShowTime Hub via an API.

2.3 User Characteristics

Users include the general customers, who need simple interfaces; customers with needs such as the option for larger text, screen reader, and seat options. Administration needs access to pricing, movie scheduling, and has the ability to read sales reports.

2.4 General Constraints

Constraints would include a system that is usable on web browsers and kiosks; processing payments must follow a guideline to reduce the risk of fraud. The system must be active and available throughout all theatre hours.

2.5 Assumptions and Dependencies

SRS assumes partner theatres to provide data through the ShowTime Hub API if they want movies to be displayed or include showtime data. Assuming that there is reliable internet access to the theatres, an addition to a third-party payment process, and loyalty points/rewards program are handled by the administration.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The website will have a web-based interface that works on both desktop and mobile devices. Users will be able to browse movies, check showtimes, choose seats, buy tickets, and order food. The interface that users can interact with will use clear navigation and readable/accessible text, so users can easily move through the different sections of the website.

3.1.2 Hardware Interfaces

The website does not need any special hardware to be accessed. Users can browse using a laptop, desktop computer, tablet, or mobile device with a valid internet connection.

3.1.3 Software Interfaces

The system will connect to participating theater systems to send and receive information like movies, showtimes, ticket prices, and availability. The website may also use outside services for receiving payments. The exact services and theater system APIs have not yet been decided.

3.1.4 Communications Interfaces

The system will use an internet connection to communicate with the website, participating theaters, and other outside services. All information related to users, purchases, etc will be transferred through secure connections.

3.2 Functional Requirements

3.2.1 <Ticket Purchasing>

3.2.1.1 Introduction

The system will allow users to purchase tickets for available movie showtimes.

3.2.1.2 Inputs

The user will select a movie then theater, then showtime, then ticket quantity, and then available seat or seats. The user will also provide or select an already existing payment method to purchase the tickets.

3.2.1.3 Processing

The system will check any selected seats to verify availability and then calculate the total purchase cost. The payment will be sent to a payment service (like Stripe) for approval. Once processed, the order will be marked complete, and the seats will be marked unavailable.

3.2.1.4 Outputs

The system will display a purchase confirmation and generate a unique ticket ID or code for each purchased ticket.

3.2.1.5 Error Handling

If a payment is declined, the system will let the user know and the purchase will not be completed. The seat selection is saved if the user wants to retry. If a selected seat is no longer available, then the user will be asked to choose another seat.

3.2.2 <Concession (Food and Drinks) Purchase>

3.2.2.1 Introduction

The system will allow users to purchase available food and drink items from participating theaters.

3.2.2.2 Inputs

- The user will select a theater location, then available concession items, then quantities, and a payment method.
- This customer selection of the “Now Showing” or “Coming Soon” category.
- The customer’s selection of a movie.
- Movie information maintained by authorized administrators or supplied through an integrated movie-information service.

3.2.2.3 Processing

The system will calculate the total cost of the order and send the payment for approval. Once approved, the order will be confirmed and recorded.

3.2.2.4 Outputs

The system will display an order confirmation and generate a unique claim code that can be used to pick up an order once at location.

3.2.1.5 Error Handling

If the payment is unsuccessful, the order will not be completed. Items are saved in the card to be retried with different payment methods. If the item is out of stock, the user will be notified.

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3.3 Use Cases

3.3.1 Use Case #1: Purchase Tickets

Use Case ID	UC-01
Name	Purchase Tickets
Priority	High
Related Requirements	3.2.1 Ticket Purchasing; 3.5.2 Reliability; Classes: Customer, PaymentMethod

Primary Actor	Customer
Secondary Actors	Payment Service (external, e.g., Stripe); Theater System (external, supplies showtime and seat availability data through the ShowTime Hub API)
Description	A customer selects a movie showtime and one or more seats, pays for them, and receives an e-ticket with a unique identifier for each seat purchased.
Preconditions	1. The customer has an internet connection and a supported web browser. 2. The showtime is scheduled in the future and has at least one available seat. 3. The Payment Service and the Theater System are reachable. 4. The customer has either an existing payment method saved to their account or a payment method to enter at checkout.
Trigger	The customer selects a showtime and chooses to buy tickets.
Main Flow	1. The customer selects a movie, a theater, and a showtime (optionally filtering by viewing format). 2. The system displays the seat map for that showtime, showing available seats, unavailable seats, and accessible seating. 3. The customer selects the ticket quantity and the desired seat(s). 4. The system verifies that every selected seat is still available for that showtime and places a temporary hold on them. The hold lasts 10 minutes and the remaining time is shown to the customer. 5. The system displays an order summary with the price of each ticket, any fees and taxes, and the final total. 6. The customer selects a saved payment method or enters a new one, and confirms the purchase. 7. The system sends a payment request to the Payment Service, tagged with a unique order reference. 8. The Payment Service approves the payment. 9. The system converts the seat holds into sold seats for that showtime, records the order as complete, and generates an e-ticket with a unique ID for each seat. 10. The system displays a purchase confirmation and makes the e-tickets available to print or to show on a mobile device.
Alternate and Error Flows	A1 - Seat unavailable (step 4): The system tells the customer which seat(s) are no longer available and refreshes the seat map. The customer chooses different seats and the flow resumes at step 3. If no seats remain, the customer is returned to showtime selection. A2 - Hold expires (steps 4 to 7): The system releases the held seats and notifies the customer. The customer returns to the seat map and must select seats again; availability is re-checked and a new hold is placed. A3 - Payment declined (step 8): The system notifies the customer that the payment was declined and no tickets are issued. While the hold is active, the customer may retry with a different payment method (resume at step 6). Otherwise A2 applies. A4 - Payment result uncertain or timed out (step 8): The system does not issue tickets and marks the order as pending. It asks the Payment Service for the final result and tells the customer not to resubmit. If payment is confirmed, the flow resumes at step 9. If payment is not confirmed, the seats are released and the customer is notified. A5 - Duplicate submission (steps 6 to 8): Repeated clicks, retries, or duplicate payment notifications for the same order reference result in one order only. The customer is never charged twice and no duplicate tickets are created.

	A6 - Customer cancels (steps 3 to 6): The system releases any held seats and ends the flow without creating an order. A7 - Theater System unreachable (steps 2 to 4): The system cannot verify seat availability, so it does not allow the purchase to proceed and tells the customer to try again later.
Postconditions (Success)	The order is recorded as complete and the payment is recorded against it. The purchased seats are marked unavailable for that showtime only. Each ticket has a unique ID and is available to the customer as an e-ticket.
Postconditions (Failure)	No valid ticket is issued and no seat is marked sold. Held seats are released. The status of the attempt (declined, expired, cancelled, or pending) is retained for reconciliation and troubleshooting.
Business Rules	Seat availability is tracked per showtime, not per physical seat. Tickets are not issued until the Payment Service confirms payment success. A seat that is sold or held for a showtime cannot be selected by another customer for that showtime. Accessible seats and companion seats are shown on the seat map and can be selected like any other seat.

3.3.2 Use Case #2: Purchase and Collect Concessions

Use Case ID	UC-02
Name	Purchase and Collect Concessions
Priority	High
Related Requirements	3.2.2 Concession (Food and Drinks) Purchase; 3.5.2 Reliability; Classes: Customer, PaymentMethod
Primary Actor	Customer
Secondary Actors	Payment Service (external); Concession Staff (validate the claim code and hand over the order at the theater)
Description	A customer orders food and drinks from a participating theater, pays online, and receives a claim code. The customer later presents the claim code at the theater to collect the order.
Preconditions	1. The customer has an internet connection and a supported web browser. 2. The selected theater participates in concession ordering and has a published concession menu. 3. The Payment Service is reachable. 4. The customer has either an existing payment method saved to their account or a payment method to enter at checkout.
Trigger	The customer chooses to order concessions.
Main Flow	1. The customer selects a theater location. 2. The system displays that theater's concession menu with item prices. Items that are out of stock are shown as unavailable or hidden. 3. The customer adds items and quantities to the order. 4. The system verifies stock for each item and calculates the subtotal, taxes, and total.

	5. The customer reviews the order, selects a saved payment method or enters a new one, and confirms. 6. The system sends a payment request to the Payment Service, tagged with a unique order reference. 7. The Payment Service approves the payment. 8. The system confirms and records the order, updates the theater's stock for the ordered items, and generates a unique claim code. 9. The system displays an order confirmation with the claim code. 10. At the theater, the customer presents the claim code to Concession Staff. 11. Concession Staff enter the claim code. The system confirms that the code is valid, belongs to that theater, and has not been used, then displays the order contents. 12. The system marks the claim code as redeemed and Concession Staff hand over the order.
Alternate and Error Flows	A1 - Item out of stock (steps 3 to 6): The system notifies the customer which item is unavailable. The customer removes it or picks a different item, and the flow resumes at step 3. Stock is checked again just before the payment request is sent. A2 - Payment declined (step 7): The system notifies the customer and no order or claim code is issued. The items stay in the customer's cart so the customer can retry with a different payment method (resume at step 5). A3 - Payment result uncertain or timed out (step 7): The system does not issue a claim code and marks the order as pending. It asks the Payment Service for the final result and tells the customer not to resubmit. If payment is confirmed, the flow resumes at step 8. If not, the customer is notified that the order was not placed. A4 - Duplicate submission (steps 5 to 7): Repeated clicks, retries, or duplicate payment notifications for the same order reference result in one order and one claim code. The customer is never charged twice. A5 - Customer cancels (steps 3 to 5): The system discards the cart and ends the flow without creating an order. A6 - Claim code invalid, already used, or from another theater (step 11): The system rejects the code, shows the reason to Concession Staff, and releases no items. A code that has been redeemed cannot be redeemed again.
Postconditions (Success)	The order is recorded as complete and the payment is recorded against it. A unique claim code exists for the order. After collection, it is marked redeemed and cannot be used again. Stock levels for the theater reflect the items ordered.
Postconditions (Failure)	No claim code is issued and no items are released. The status of the attempt (declined, cancelled, or pending) is retained for reconciliation and troubleshooting.
Business Rules	A claim code is not issued until the Payment Service confirms payment success. Concession menus, prices, and stock are specific to each theater location. A claim code can be redeemed once, and only at the theater where the order was placed.

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3.4 Classes / Objects

Core classes and objects for ShowTime Hub, are taken from user inputs and the concept of the system described in Section 2.

3.4.1 Customer

3.4.1.1 Attributes

name, savedPaymentMethods, customerID, badgeList, passwordHash

3.4.1.2 Functions

login(), viewPurchaseHistory(), register()

Creating and Managing user account

3.4.2 PaymentMethod

3.4.2.1 Attributes

customerID, paymentMethodID, lastFourDigits, cardType, expirationDate

3.4.2.2 Functions

addPaymentMethod(), chargePaymentMethod(), removePaymentMethod()

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3.5 Non-Functional Requirements

3.5.1 Performance

With normal traffic, at least 95% of pages and movie/showtime searches should load within 3 seconds. Seat availability should be updated within 3 seconds after the system receives new information from the theater service.

3.5.2 Reliability

The system will not issue any tickets or codes for concessions until external service confirms that payment has been successful. Seats already sold will not be sold to other users. Failed transactions should not have any order ids or other types of records created in the system.

3.5.3 Availability

Website is expected to run 99% of the time, not including maintenance or outages caused by issues outside of our reach. Scheduled maintenance should be performed during low traffic periods if possible.

3.5.4 Security

All account and payment method related information will be transmitted through secure connections. User passwords and personal data will be stored securely and not as plain text. Admin features are only accessible to authorized admin accounts.

3.5.5 Maintainability

The system should be created and organized such that changes to one major feature like theater integration or payment service provider, do not require major changes to unrelated parts of the website. Any errors should be logged with enough information to identify and fix the issues.

<Showtime Hub>

3.5.6 Portability

The website will support current versions of most major web browsers, including Chrome, Edge, Firefox, Safari, etc. The interface should work on both desktop and mobile screen sizes without removing important text and functionality.